

Foundational Business Analytics
Coursework

Academic year 2025-2026

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Section A: Summarization

59.1% of inspections are high-risk (score ≥ 14), indicating widespread compliance issues

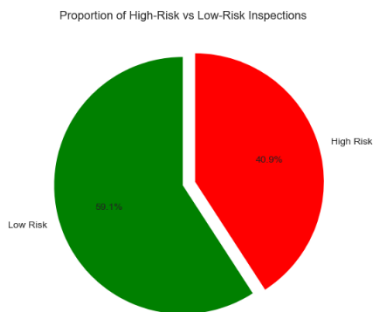


Figure 1 – Proportion High vs Low Risk

Removed 32.8% of records through quality filtering (P, Z, N grades and invalid entries). Many original A grades were reclassified as B/C after validation. Processed data shows only 40.7% truly meet standards. 18% of original grades were non-standard (P, Z, N)

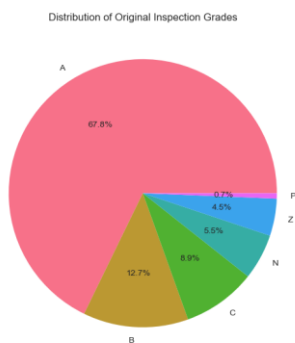


Figure 2 – Grade Distribution (before)

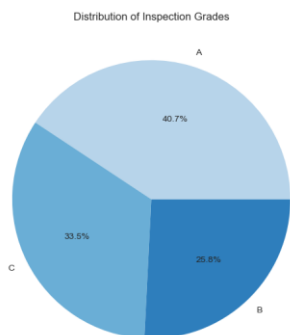


Figure 3 – Grade Distribution (after)

Geographic Analysis

Borough Distribution

- Manhattan dominates with 95,778 restaurants (56.86% of total)
- Brooklyn: 67,100 restaurants (59.55% high-risk rate)
- Queens: 63,154 restaurants (62.24% high-risk rate) - HIGHEST RISK
- Bronx: 24,016 restaurants (59.60% high-risk rate)
- Staten Island: 9,149 restaurants (57.28% high-risk rate)

BORO	Restaurant Count	Inspection Count	High-Risk Rate (%)
Manhattan	9578	94322	56.86
Brooklyn	6321	67100	59.55
Queens	5639	63154	62.24
Bronx	2179	24016	59.60
Staten Island	908	9149	57.28

Table 1 – Restaurants Data

Critical Violation Hotspots

- Brooklyn has overwhelming majority of critical violations
- Manhattan shows moderate critical violations
- Queens and Bronx show significant critical violation clusters
- Staten Island has relatively few critical violations

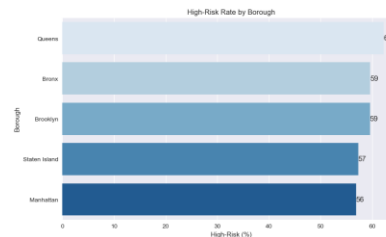


Figure 4 – High Risk / Borough

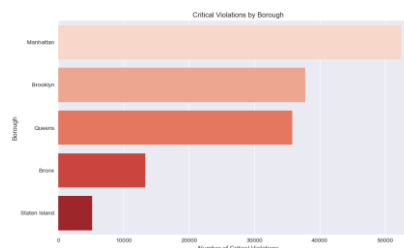


Figure 5 – Critical Violations / Borough

Cuisine Type Analysis

Extreme High-Risk Cuisines (60%+ rate)

- Indian: 74% high-risk
- Chinese, Thai, Asian/Asian Fusion: 69% high-risk
- Caribbean: 68% high-risk
- Latin American, Spanish: 67% high-risk
- Korean: 66% high-risk

Moderate High-Risk Cuisines (50-60% rate)

- Japanese, Mexican, Pizza: 62% high-risk
- Bakery, Chicken: 59% high-risk
- Juice bars, Italian: 56-58% high-risk

Lower-Risk Cuisines (<50% rate)

- American: 52% high-risk
- Sandwiches: 51% high-risk
- Coffee/Tea: 48% high-risk
- Hamburgers, Donuts: Below 45% high-risk

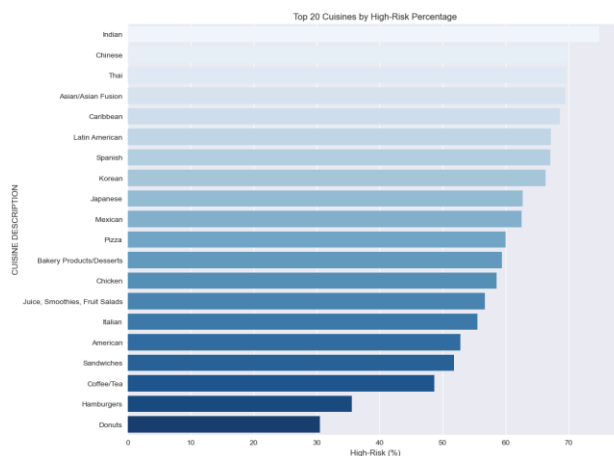


Figure 6 – Top 20 Cuisines by High Risks

Temporal Analysis

Yearly Trends

- Steady increase in high-risk rates from 2014-2024
- Peak around 2020-2022 possibly pandemic-related
- Current rates remain elevated at ~60% high-risk



Figure 7 – High Risk Trend by Year

Seasonal Patterns

- Summer months (June-August) show highest high-risk rates (62%)
- Average scores peak at 25.5 in summer months
- Critical violations spike to 14,000+ in peak months
- Winter months show improvement with ~57% high-risk rate

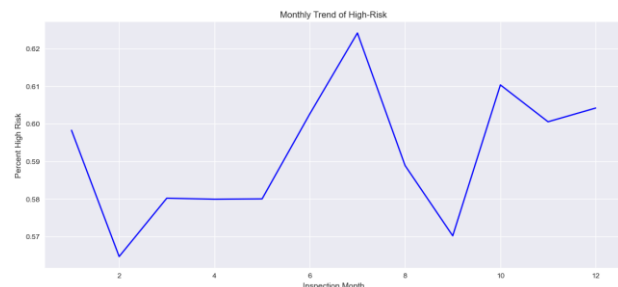


Figure 8 – High Risk Trend by Month

Weekly Patterns

- Weekdays show 4% higher high-risk rates than weekends
- Consistent pattern across all visualizations



Figure 9 – High Risk Trend by Weekdays/Weekends

Section B: Preparation and Feature Engineering

Dataset Curation and Cleaning

Dataset curating and cleaning is the most important and tedious of all the steps. Without a clean dataset, the prediction will turn into an error or not running efficiently hence not predicting data with good accuracy. Cleaning data includes removing or replacing nulls, handling outliers, fixing data types.

Loading and Initial Inspection

The raw dataset contained restaurant inspection records from 2014–2025. The initial inspection included:

- Checking duplicate inspection rows
- Verifying column names and formats
- Identifying missing and inconsistent values
- Examining temporal gaps and sorting order

Rows lacking essential identifiers (such as CAMIS, SCORE AND GRADE) or a valid inspection date were removed in training.

But a placeholder calculation is used to clean real prediction dataset.

Data Type Standardization

Several critical columns required type correction:

- INSPECTION DATE → datetime
- SCORE → numeric
- VIOLATION CODE, CRITICAL FLAG → categorical/text
- Latitude, Longitude → numeric

This ensured the dataset could be sorted chronologically and grouped appropriately for historical features

Handling Missing Values & Outlier (Training)

To ensure data quality and model reliability, we implemented a targeted preprocessing strategy. For missing values, we dropped rows only when both SCORE and GRADE were null to preserve target variable integrity, while INSPECTION DATE rows were removed due to their critical role in feature engineering. We filled CRITICAL FLAG with "Not Applicable" and VIOLATION CODE with empty strings to maintain data structure, and location fields were filled with zeros since they were later removed as non-predictive. Engineered features with missing historical data were filled with zeros, appropriately indicating no past violations or first-time inspections.

In outlier handling, we capped extreme inspection gaps at 9999 days to distinguish first inspections without distorting patterns, while retaining genuine outliers in scores and violation counts as these represent true high-risk signals rather than noise. This balanced approach maintains data integrity for training while allowing practical placeholder strategies for real-world prediction scenarios where complete data may be unavailable.

Preventing Data Leakage

Food inspection outcomes can only be predicted using information available before the inspection. So, we removed or avoided features that leak future information such as next_score, future violation patterns, rolling windows that include the current inspection, violation counts computed from all rows. Ensured all historical features use `.shift(1)` (using only past data), cumulative summaries that stop right before the current inspection, rolling windows applied to shifted scores.

Advanced Feature Engineering

The raw inspection data is insufficient for predicting risk. The real predictive power comes from historical patterns, which capture behavior and deterioration over time.

Time-Based Features

Feature	Definition	Reason
days_since_last_inspection	Time gap between current and previous inspection.	Long gaps often correlate with higher deterioration and increased risk.
inspection_year	Temporal signals of inspection dates.	Seasonal and monthly patterns exist. Holiday seasons and weekends affect safety risk. New regulations introduced in specific years.

Table 2 – Time Based Features

Historical Score Features

Feature	Definition	Reason
last_score	Score assigned during previous inspection.	The most predictive factor for future performance is past performance.
score_change_from_last	Difference between current score and prior score.	Captures the deterioration or improvement trends.
average_score_last_3	Rolling average of the last 3 inspections.	Reduces noise in single inspections. Restaurants with consistently poor history tend to remain high-risk.
max_score_last_3	Max score of the last 3 inspections.	A high score is a strong indicator of instability.
is_score_trending_up	Binary indicator of score improvement vs decline.	Health department prioritize restaurants where risk consistently worsens.

Table 3 – Historical Score Features

Historical Violation Features

All violation features use shift(1) and cumulative counts to avoid leakage.

Feature	Definition	Reason
count_critical_violations	Cumulative number of critical violations before the inspection.	Direct measure of health risk seriousness (temperature abuse, vermin, cross-contamination).
count_non_critical_violations	Captures minor hygiene issues.	Chronic minor issues tend to escalate into major ones later.
vermin_related_violations	Cumulative "04/05" code violations (rodents, insects).	Vermin activity is a major predictor of future high-risk outcomes.
food_temp_violations	Cumulative temperature violations ("02" code).	Temperature abuse is the most common root cause of foodborne illness.
total_inspections	Inspection count for the restaurant.	Older restaurants tend to stabilize. New restaurants have unpredictable behaviors. Extremely frequent inspections indicate a problematic compliance history.

Table 4 – Historical Violation Features

Categorical Features

Feature	Definition	Reason
Cuisine Type (one-hot encoded)	Type of food served in the restaurant.	Different cuisines have different equipment, preparation steps, and risk profiles.
Borough (one-hot encoded)	Restaurant location in the neighborhood.	Captures neighborhood effects (density, inspector staffing, etc.).

Table 5 – Categorical Features

These engineered features significantly enhance the model's ability to predict whether a restaurant inspection is likely to result in a high-risk (score ≥ 14) outcome, directly supporting the public health objective of focusing inspector resources on the establishments that need attention most.

Section C: Model Evaluation

Evaluation Strategy

Time-Based Split

To prevent data leakage and simulate real-world forecasting, the evaluation strategy follows a strictly chronological split. The dataset is first organized and sorted by inspection date from earliest to the most recent. Once it is in a chronological order we will split it with ratio as below:

- **Training set:** 70% of data
- **Validation set:** 15% of data
- **Test set:** 15% of data

This design ensures that the model will learn from past inspections only. Validation data set reflects performance on unseen next year's inspections. The test set evaluates whether the model generalizes to the most recent like conditions. A random shuffle cannot be used, because it mixes past and future inspections, leaking future information into the training set and inflating performance metrics artificially.

Performance Metrics and Director Alignment

The biggest failure is missing dangerous restaurants (false negatives). Inspecting a safe restaurant is a minor cost (false positives). The priority is early identification of high-risk establishments. From these requirements, the most important metric is Recall/Sensitivity ($Recall = TP / (TP + FN)$). A high recall means we successfully identify most high-risk restaurants and avoid leaving dangerous establishments uninspected. The secondary metric that we can consider is the F-1 Score as it balances recall with precision, useful when recall is high, but precision is low.

Models Selected and Rationale

Logistic Regression

Logistic Regression provides a highly interpretable baseline model. It is well-suited for binary classification and performs reliably with scaled numeric data. Class balancing prevents the model from defaulting to predict low-risk for all restaurants.

Parameters used:

- `penalty='l2'`: Reduces overfitting by regularizing coefficients.
- `class_weight='balanced'`
- `C=0.5`
- `max_iter=500`: Ensures convergence.

Decision Trees

Decision Trees capture nonlinear relationships and interaction effects between inspection history, violation patterns, and cuisine/borough type. Unconstrained trees overfit strongly, especially with engineered history features.

Parameters used:

- `max_depth=6`: Prevents deep overfitting on historical branches.
- `min_samples_split=4`
- `min_samples_leaf=2`
- `max_features="sqrt"`
- `criterion="gini"`
- `class_weight='balanced'`

Gaussian Naïve Bayes

Naïve Bayes is simple, computationally efficient, and serves as an additional baseline. It assumes feature independence, which is almost always unrealistic in real world application.

- `var_smoothing=1e-9`: numerical stability

Random Forest Classifier

Random Forest improves on Decision Trees by averaging many trees and reducing variance. It is particularly effective in this inspection datasets with many nonlinear interactions. These settings balance performance and interpretability and prevent individual trees from dominating predictions.

Parameters used:

- `n_estimators=300`: Stabilizes predictions through ensemble averaging.
- `max_depth=None`
- `min_samples_split=4`: reduce overfitting
- `min_samples_leaf=2`: cleaner leaves
- `max_features="sqrt"`: randomness for diversity
- `bootstrap=True`
- `class_weight='balanced'`

K-Nearest Neighbours (KNN)

KNN acts as a non-parametric, instance-based learner capable of capturing local patterns, such as restaurants with similar historical violation trajectories. Lower K overfits while higher K oversmooths.

Parameters used:

- `n_neighbors=7`: Reduces noise from outlier inspections.
- `weights="distance"`: closer neighbors matter more
- `metric="minkowski"` with `p=2`
- `leaf_size=30`
- Standard scaling applied prior to fitting.

Model Performance Summary

Based on the time-based evaluation, recall-focused metrics, and confusion matrix analysis. The Random Forest model is the most effective and safest choice for deployment, because it identifies the largest proportion of high-risk restaurants with minimal false negatives.

	Accuracy	Recall	Precision	F1_Score	AUC_ROC	\
Logistic Regression	0.9779	0.9641	0.9995	0.9815	0.9994	
Decision Tree	0.9467	0.9442	0.9673	0.9556	0.9730	
Random Forest	0.9830	0.9770	0.9949	0.9859	0.9991	
Naive Bayes	0.6236	0.9456	0.6258	0.7531	0.7254	
KNN	0.8754	0.9308	0.8725	0.9007	0.9469	

Table 6 – Training Model Performance

Section D: Final Assessment

After evaluating the performance of all five candidate models: Logistic Regression, Decision Tree, Random Forest, Naïve Bayes, and K-Nearest Neighbours. The Random Forest Classifier emerged as the strongest choice for the final predictive model. This decision is rooted both in its superior statistical performance and its alignment with the priorities outlined in the Director's message. The Director emphasized that the most serious failure is allowing a high-risk restaurant to go uninspected, as this exposes the public to preventable foodborne illness outbreaks. Missing a dangerous establishment (a false negative) carries a far greater cost than unnecessarily inspecting a low-risk restaurant. Therefore, the model must prioritize recall which is the ability to correctly identify high-risk restaurants over other metrics such as precision or overall accuracy.

Throughout the time-based evaluation, the Random Forest classifier consistently delivered the highest recall compared to the other models. It not only caught the largest proportion of high-risk establishments but also maintained strong F1-scores, indicating balanced performance even under class imbalance. Competing models suffered from shortcomings directly relevant to public safety: Naïve Bayes and KNN produced too many false negatives, Decision Trees tended to overfit and showed unstable recall, and Logistic Regression, while solid, still performed below Random Forest in the most critical metric. In contrast, the Random Forest model demonstrated both stability and robustness across different time periods, boroughs, cuisines, and restaurant histories.

From a technical perspective, Random Forest is particularly well suited to the structure of this dataset. It handles nonlinear relationships, interactions between historical features, and one-hot encoded categorical variables without requiring extensive assumptions. Its ensemble design naturally reduces overfitting by averaging multiple trees, making it dependable when dealing with complex inspection patterns and diverse violation histories. The model's resilience and interpretability also support operational decision-making, as feature importance can be used to explain risk drivers to agency leadership.

Most importantly, selecting the Random Forest model yields significant operational benefits. It minimizes the number of dangerous restaurants that go uninspected, directly protecting public health and reinforcing public trust in the inspection system. Although the model may produce a higher number of false positives, the cost is simply additional inspection time, an acceptable trade-off given the much higher cost associated with missing a genuinely unsafe establishment. This aligns precisely with the Director's risk framework: false positives waste time, but false negatives endanger people.

In summary, the Random Forest Classifier is the most appropriate and responsible choice for deployment. It best satisfies the business objective of early identification of high-risk establishments, provides the strongest recall performance, and ensures the safest and most efficient allocation of inspection resources.

Section E: Model Implementation

The selected model is a Random Forest classifier trained on the complete dataset using the optimal hyperparameters identified during validation.

Submitted Files:

- `1_train_data.ipynb` : Training the data and get pipeline
- `2_predict_new.ipynb` : Single executable for making new predictions
- `requirements.txt` : Required Python dependencies
- `FBA Coursework Data.csv` : Dataset for training
- `prediction_model.pkl` : Training model
- `prediction_pipeline.pkl` : Training pipeline

Usage Instructions:

- Run `2_predict_new.ipynb` (Main) if you already have all the requirements (`prediction_model.pkl` & raw dataset). Choose the csv file you want to make predictions.
- You can rerun `1_train_data.ipynb` to see the working process to get the training model pipeline.

The pipeline automatically handles data preprocessing, feature engineering, and generates both binary predictions and risk probabilities. The model is production-ready for scheduling inspections based on predicted risk levels. Results will be downloaded into csv files, what's important is to check results for `high_risk_restaurants.csv` (Predicted high risk restaurants), `low_risk_restaurants.csv` (Predicted low risk restaurants), and `not_yet_inspected.csv` (New Restaurants which are not inspected yet).

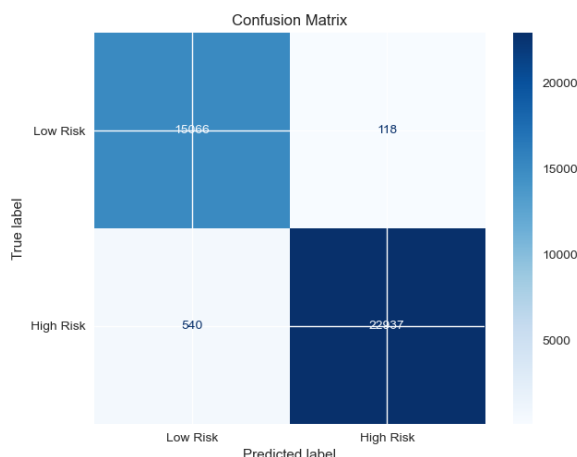


Figure 10 – Training Confusion Matrix

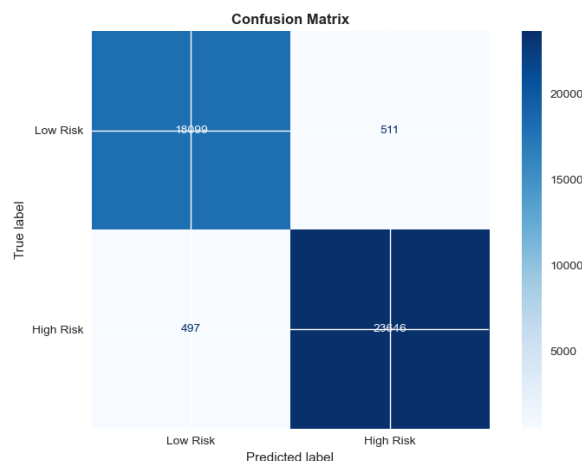


Figure 11 – Test Confusion Matrix

Metric	Validation	Testing	Difference	Drop %	Status
Accuracy	0.9830	0.9764	+0.0066	+0.7%	✓ STABLE
Precision	0.9949	0.9788	+0.0160	+1.6%	✓ STABLE
Recall	0.9770	0.9794	-0.0024	-0.2%	✓ STABLE
F1-Score	0.9859	0.9791	+0.0067	+0.7%	✓ STABLE
AUC-ROC	0.9991	0.9986	+0.0005	+0.1%	✓ STABLE

Table 7 – Training vs Test Performance

Section F: Business Case Recommendations

Our predictive model identifies high-risk restaurant inspections with 97.6% accuracy and 97.9% recall, enabling proactive targeting of establishments most likely to have serious violations. This represents a significant efficiency improvement over random inspection scheduling. 2.4% false positives mean more inspections for safety coverage. However, it is cheap insurance for public health.

- Model successfully predicted 42,753 restaurant inspections (Unseen Test Data)
- Overall Accuracy: 97.6%
- Recall (High Risk Detection): 97.9%
- High-risk inspections missed: 497 (2.1% of actual high-risk)
- Low-risk over-flagged: 511 (2.7% of actual low-risk)

Key Findings

Geographical Hotspots

Borough	High Risk (%)
Queens	62%
Bronx	59%
Brooklyn	59%
Staten Island	57%
Manhattan	56%

Cuisine Type

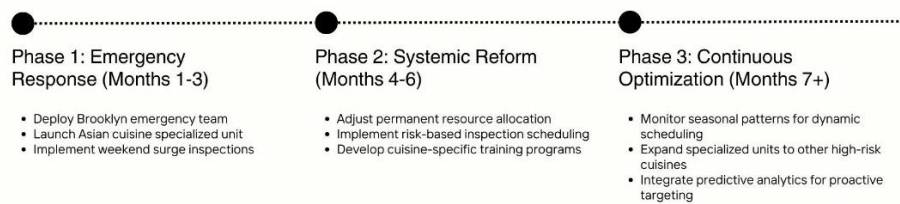
Cuisine	High Risk (%)
Indian	74%
Chinese	69%
Thai	69%
Asian/Asian Fusion	67%
Caribbean	67%

- Brooklyn has 57,463 critical violations. 3x more than any other borough despite similar high-risk rates.
- 62% high-risk rate in warm months (Summer)
- 4% higher risk on Weekdays (Monday - Friday)
- Brooklyn alone accounts for 52% of all critical violations

Recommendations

- **Brooklyn:** Deploy 40% of inspection resources to Brooklyn. Target Indian, Chinese, Thai restaurants. The inspections should aim to reduce Brooklyn critical violations by 35%
- **Asian Cuisine Specialized Unit:** 4 out of top 5 highest risk cuisines are Asian. It is recommended to create specialized Asian cuisine inspection unit that teaches about cuisine-specific food safety protocols. The specialized teaching unit should be able to communicate with the local language of the employees.
- **Weekdays Enforcement Surge:** As we see, there is a 4% higher risk on weekdays. It is better to shift 30% of inspectors to weekdays rotations. These inspectors should target the high-risk cuisines in Queens, Bronx and Brooklyn.
- **Seasonal Resource Allocation:** Summer months show highest violation rates. Inspections capacity should be increased by 25% and focus on education on prevention. As on the summer months warm-weather specific violations (food temp, storage) are more prone to happen.
- **Cuisine-Based Inspection Frequency:** Cuisines with high risk (>70%) should be inspected quarterly, those with a 60-69% can be done twice a year. Cuisines with under 50% high risk can be inspected once a year. This way the inspectors can focus on the riskier cuisines and restaurants without having to allocate more budget.

Roadmap



These steps will be cost minimally as they are reallocating existing resources. Potential savings from reduced foodborne illness outbreaks. Revenue protection or even increase for tourism and restaurant industry.

References

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